



UNIVERSITY OF WESTERN AUSTRALIA

MASTER OF PHYSICS (COMPUTATIONAL PHYSICS)

**Quantum Circuits for Solving Differential Equations via
Physics-Informed Effective Hamiltonians and GQSP
Quantum Imaginary Time Evolution**

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Supervised by

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14 September 2026

Declaration

This is to certify that:

1. This thesis comprises of my original work.
2. Due acknowledgment has been made in the text to all other materials used.
3. This thesis consists of less than 60 pages inclusive of tables and figures but exclusive of references and appendices.

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Summary of Student Achievement

Abstract

Brief summary.....

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Notation and Conventions

A quick reference for symbols used across the thesis; each is properly introduced at its first use in the text below.

Discretisation

Symbol	Meaning
n	Number of qubits in the Chebyshev coefficient register (the resolution).
$N = 2^n$	Chebyshev truncation degree; dimension of the single-field Hilbert space.

Differential-equation structure

Symbol	Meaning
k	Order of an ordinary differential equation (ODE).
p	Polynomial (nonlinear) degree of the ODE (§4).

Chebyshev and descriptor operators

Symbol	Meaning
T_ℓ	Chebyshev polynomial of the first kind, degree ℓ .
G	Dense $N \times N$ Chebyshev differentiation matrix (the collocation route’s operator; Wu et al. [1]).
$\tilde{B}, \tilde{D}$	Banded factors of G (Appendix B).
M	Coefficient-multiplication operator in Chebyshev coefficient space (Appendix B).
w	The stacked descriptor state, $w = (c^{(0)}; \dots; c^{(k)})$.
A	The residual operator of a linear-system reformulation of the ODE (dense in the collocation route, banded in the descriptor route).

Physical and complexity quantities

Symbol	Meaning
H	The physics-informed effective Hamiltonian (Wu et al. [1]).
α	Block-encoding subnormalisation factor.
Δ	Spectral gap of H between its ground state and first excited state.
ϵ_G	Block-encoding / truncation error associated with representing G .

1 Introduction

1.1 Differential Equations as a Target for Quantum Computation

1.2 The Encoding Wall in Physics-Informed Quantum Solvers

1.3 Contributions and Scope

1.4 Outline

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2 Background and Literature Review

2.1 Quantum Computation Preliminaries

In classical computing, the fundamental unit of information is the bit, which can take one of two values: 0 or 1.

The fundamental unit of information in quantum computing is the qubit, which can exist in a superposition of states, allowing it to encode both 0 and 1 simultaneously. This property enables quantum computers to perform certain computations more efficiently than classical computers.

Definition 1 (State Space): The set of all quantum states that a quantum system can occupy is a complex vector space with inner product structure known as a Hilbert space (denoted $\mathcal{H}$). A finite dimensional Hilbert space is isomorphic to $\mathbb{C}^N$ ($\mathcal{H} \cong \mathbb{C}^N$) for some $N \in \mathbb{N}$, where N is the dimension of the Hilbert space. A quantum state is a unit vector in this Hilbert space, represented in dirac notation as $|\psi\rangle \in \mathcal{H}$, with $\langle\psi|\psi\rangle = 1$ (normalisation condition). A quantum state $|\psi\rangle$ can be expressed as a linear combination of basis states $\{|i\rangle\}_{i=0}^{N-1}$, where $|i\rangle$ is the i^{th} basis state of the Hilbert space.

$$|\psi\rangle = \sum_{i=0}^{N-1} \psi_i |i\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{N-1} \end{pmatrix} \quad (2.1)$$

Its hermitian conjugate (or adjoint) is denoted as $\langle\psi| = |\psi\rangle^\dagger$. Note that for some $e^{i\theta}$ for $\theta \in [0, 2\pi)$, the states $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are physically indistinguishable, and thus correspond to the same physical quantum state. This is known as a global phase factor.

Definition 2 (The Qubit): A quantum bit (qubit) is a two-state quantum system, characterised by $|\psi\rangle \in \mathbb{C}^2$.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (2.2)$$

where $\alpha, \beta \in \mathbb{C}$, $|0\rangle$ and $|1\rangle$ are the computational basis states, and $|\alpha|^2, |\beta|^2$ are the probabilities of measuring the qubit in the $|0\rangle$ and $|1\rangle$ states, respectively. The normalisation condition requires that $|\alpha|^2 + |\beta|^2 = 1$.

Definition 3 (Composite Systems): The state space of a composite quantum system is the tensor product of the state spaces of its subsystems. Consider two quantum states $|\psi\rangle_A \in \mathcal{H}_A$

and $|\psi\rangle_B \in \mathcal{H}_B$, the composite state $|\psi\rangle_{AB}$ is given by:

$$|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\psi\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B \quad (2.3)$$

where $\otimes$ denotes the tensor product and is often hidden by convention. The dimension of the composite state space is the product of the dimensions of the individual state spaces. If a composite system cannot be written as a tensor product of its subsystems, it is said to be entangled. Entanglement is a unique feature of quantum mechanics that has no classical counterpart and plays a crucial role in quantum information processing.

We often consider the state of a composite system of n qubits, which is represented in a 2^n -dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^{2^n}$.

2.1.1 Quantum Gates and Circuits

Definition 4 (Quantum Algorithm): A quantum algorithm is a sequence of operations that manipulates qubits to perform a specific computation.

The goal of any quantum algorithm is to manipulate quantum states in a controlled manner to achieve a desired outcome. This is accomplished through the application of quantum gates, which are unitary operations that transform quantum states. Quantum gates can be combined to form quantum circuits, which represent the sequence of operations performed on qubits. Each quantum gate can be implemented using physical operations on hardware, and the design of efficient quantum algorithms is a central challenge in quantum computing research.

Definition 5 (Unitary Evolution): The evolution of a closed quantum system between two states is described by unitary transformations $U \in \mathbb{C}^{2^n \times 2^n}$, where n is the number of qubits in the system.

$$|\psi\rangle_{final} = U |\psi\rangle_{initial}, \quad U^\dagger U = U U^\dagger = I \quad (2.4)$$

where $U^\dagger$ is the hermitian conjugate of U and I is the identity operator. Unitary transformations preserve the inner product and thus the norm of quantum states, ensuring that states have a physical interpretation as probability amplitudes.

These unitary transformations are known as quantum gates, and each gate serves a specific purpose in manipulating quantum states.

The quantum circuit diagram offers a succinct and intuitive representation of quantum algorithms, analogous to Feynman diagrams in quantum field theory.

Time flows left to right, and each wire (horizontal line) represents a qubit or a quantum register. Gates are placed sequentially on the wires indicating operations applied to qubits. An example is the Hadamard gate, which creates superposition, and the CNOT gate, which entangles qubits. A full list of quantum gates and their corresponding circuit symbols can be found in .

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2.1.2 Complexity of Quantum Algorithms

Definition 6 (Gate Complexity): The total number of quantum gates used in a quantum circuit.

Definition 7 (Circuit Depth): The maximum number of gates applied sequentially on any qubit in a quantum circuit.

Definition 8 (Ancillary Qubits): Additional qubits introduced into a quantum circuit beyond the primary input qubits, typically initialized to a baseline state (such as $|0\rangle$) to assist with intermediate calculations, temporary storage, or entanglement generation.

Definition 9 (Big- $\mathcal{O}$ Notation): Big- $\mathcal{O}$ notation is a mathematical notation used to describe the asymptotic behavior of functions.

$$\mathcal{O}(f(n)) = \{g(n) : \exists c > 0, n_0 \in \mathbb{N} \text{ such that } 0 \leq g(n) \leq cf(n) \text{ for all } n \geq n_0\} \quad (2.5)$$

In the context of quantum algorithms, it is used to express the scaling of gate complexity, circuit depth, and space requirements with respect to the number of qubits n .

For example, if a quantum algorithm has a gate complexity of $\mathcal{O}(n^2)$, it means that the number of gates required grows quadratically with the number of qubits.

A quantum circuit is said to be **efficient** if its circuit depth, gate complexity, and the number of ancillary qubits all scale polynomially with the number of qubits n .

2.1.3 Noisy Intermediate-Scale Quantum (NISQ) Era

Write this section. Use Josh's as a guide.

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3.1	Overview
3.2	Design Rationale

The differentiation operator G is dense. Generally, dense operators are expensive to block-encode and sparse matrices are cheap.

The doubled Hilbert space method can create degenerate kernels. Quantum imaginary time evolution (QITE) can't select the physical solution from amongst a set of degenerate ground states, so although it is possible to encode the solution to the original problem in the ground state of the Hamiltonian, it is not possible to extract it from the ground state of the descriptor Hamiltonian.

Add a reference for this

Remember to define dense vs sparse

3.3 Descriptor Form Construction

Differentiation of a Chebyshev series is a sparse operation written densely. Let $f = \sum_{\ell} c_{\ell}^{(0)} T_{\ell}$ have derivative $f' = \sum_{\ell} c_{\ell}^{(1)} T_{\ell}$; for $\ell \geq 2$ the coefficient vectors $c^{(0)}$ and $c^{(1)}$ are tied by the recurrence

$$c_{\ell-1}^{(1)} - c_{\ell+1}^{(1)} = 2\ell c_{\ell}^{(0)}, \quad (3.1)$$

with the $\ell = 1$ row carrying an extra factor of 2 on $c_0^{(1)}$, a normalisation artefact of T_0 that Appendix B states in full and derives. Each row relates the coefficient $c_{\ell}^{(0)}$ to just two coefficients of the derivative. In the normalised Chebyshev basis this is a matrix identity

$$\tilde{B} c^{(1)} = \tilde{D} c^{(0)}, \quad (3.2)$$

with $\tilde{B}$ banded (its main diagonal and its second superdiagonal), and $\tilde{D}$ carrying a single superdiagonal, such that both hold $\mathcal{O}(N)$ non-zero entries. The dense Chebyshev derivative matrix G is the operator that sends a function's coefficients directly to those of its derivative, $c^{(1)} = G c^{(0)}$. A simple rearranging of Equation (3.2) recovers it as the inverse

$$G = \tilde{B}^{-1} \tilde{D}, \quad (3.3)$$

an identity that holds exactly. The density of G is due to inverting a banded matrix, and Equation (3.2) is the same linear map before that inverse is done. This is the discrete form of the ultraspherical spectral method [2], in which differentiation becomes sparse once the derivative is allowed to sit in a polynomial basis one order above the function.

Consider the homogeneous order- k linear ODE

$$\sum_{j=0}^k a_j(x) f^{(j)}(x) = 0, \quad x \in [-1, 1], \quad (3.4)$$

where $a_j(x)$ are the (possibly x -dependent) coefficients of each j^{th} derivative $f^{(j)}$.

We discretise by representing every function appearing in the equation by its first N coefficients in the normalised Chebyshev basis. Concretely, let $c^{(j)} \in \mathbb{C}^N$ denote the coefficient vector of $f^{(j)}$, for $j = 0, \dots, k$ and stack these into a single vector,

$$w = (c^{(0)}; c^{(1)}; \dots; c^{(k)}). \quad (3.5)$$

Two kinds of linear relations couple the blocks $c^{(j)}$:

Recurrence rows. The coefficients of consecutive derivatives are related by

$$-\tilde{D} c^{(j)} + \tilde{B} c^{(j+1)} = 0, \quad j = 0, \dots, k-1. \quad (3.6)$$

This gives k rows of the system, each enforcing the derivative relationship between $c^{(j+1)}$ and $c^{(j)}$. These rows are algebraic since they carry no derivative and only fix each $c^{(j+1)}$ as the derivative of $c^{(j)}$, being Equation (3.2) applied block-wise and hence equivalent to $c^{(j+1)} = Gc^{(j)}$ on $\ker A$.

Differential row. The differential equation itself becomes one further row, obtained by multiplying each coefficient vector $c^{(j)}$ by the (possibly x -dependent) coefficient $a_j(x)$ and summing:

$$\sum_{j=0}^k M_{a_j} c^{(j)} = 0, \quad (3.7)$$

where M_{a_j} is the matrix representing multiplication by $a_j(x)$ in coefficient space (reducing to $a_j I_N$ when a_j is constant). This single row is the equation itself: we carry the intermediate derivatives as their own unknown coefficient vectors within w , rather than representing them via $c^{(j)} = G^j c^{(0)}$, so that Equation (3.3) (the only dense step) never appears.

Together these $k + 1$ block rows (k recurrence rows and one differential row) define a linear system for w and transform the original ODE problem into a differential-algebraic equation (DAE) system.

Definition 10 (Descriptor form residual operator): Collecting all block rows, the descriptor form chebyshev residual operator A is the $(k + 1)N \times (k + 1)N$ block matrix

$$A = \begin{pmatrix} -\tilde{D} & \tilde{B} & & & \\ & -\tilde{D} & \tilde{B} & & \\ & & \ddots & \ddots & \\ & & & -\tilde{D} & \tilde{B} \\ M_{a_0} & M_{a_1} & \cdots & \cdots & M_{a_k} \end{pmatrix}, \quad (3.8)$$

where the first k rows are the recurrence rows and the $(k + 1)$ -th row is the differential row, so that the whole system is compactly written as

$$Aw = 0. \quad (3.9)$$

A first-order example. For $f' + 2f = 0$ the state is $w = (c^{(0)}; c^{(1)})$, and the two facts relating its blocks are the recurrence $\tilde{B}c^{(1)} - \tilde{D}c^{(0)} = 0$ and the equation $2c^{(0)} + c^{(1)} = 0$, giving

$$A = \begin{pmatrix} -\tilde{D} & \tilde{B} \\ 2I_N & I_N \end{pmatrix}, \quad w = \begin{pmatrix} c^{(0)} \\ c^{(1)} \end{pmatrix}. \quad (3.10)$$

Every block is $\tilde{B}$, $\tilde{D}$ or a multiple of the identity, so A is banded, against the dense $G + 2I$ that substituting $c^{(1)} = Gc^{(0)}$ would produce. The fully populated $N = 4$ instance of Equation (3.10), the entries of $\tilde{B}$ and $\tilde{D}$, and the exactness of Equation (3.3), are given in Appendix B.

Cost. The reformulation trades a higher dimensional Hilbert space (and with it a higher qubit count) for a sparser residual operator. w has $(k+1)N$ components at the classical, unpadded level; realising the field register on $\lceil \log_2(k+1) \rceil$ qubits pads this to $2^{\lceil \log_2(k+1) \rceil} N$ (Appendix B), which is what should be compared against the N of the collocation form. Each block of A is $\tilde{B}$, $\tilde{D}$, or a banded multiplication operator (bandwidth q for a degree- q polynomial coefficient; Appendix B), so A carries $\mathcal{O}(1)$ occupied diagonals per block, whereas the collocation form residual $\sum_j a_j G^j$ is dense with $\Theta(N^2)$ non-zero entries. The $c^{(0)}$ -block of the recovered null vector is the Chebyshev coefficient vector the collocation form solves for, agreeing with it to within numerical precision, quantified in Section 5. Carrying the intermediate derivatives instead of eliminating them converts the dense physics-informed residual into a banded one, and every later stage of the pipeline inherits that structure.

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A Quantum Gate Reference

B Chebyshev and Ultraspherical Identities

This appendix records the entries of the banded factors $\tilde{B}$ and $\tilde{D}$ introduced in Section 3.3, the exact factorisation $G = \tilde{B}^{-1}\tilde{D}$, and the fully populated $N = 4$ instance of the first-order descriptor operator.

B.1 The differentiation recurrence

For $f = \sum_{\ell \geq 0} a_\ell T_\ell$ with derivative $f' = \sum_{\ell \geq 0} p_\ell T_\ell$ on $[-1, 1]$, the Chebyshev coefficients satisfy

$$c_{\ell-1} p_{\ell-1} - p_{\ell+1} = 2\ell a_\ell, \quad \ell \geq 1, \quad (\text{B.1})$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$; the $\ell = 1$ row therefore carries the factor $c_0 = 2$ on p_0 . Truncating to $N = 2^n$ coefficients and dropping the p_N term, Equation (B.1) is the matrix identity $\tilde{B}_{\text{std}} p = \tilde{D}_{\text{std}} a$ with $\tilde{D}_{\text{std}}$ a single superdiagonal, $(\tilde{D}_{\text{std}})_{\ell, \ell+1} = 2(\ell + 1)$, and $\tilde{B}_{\text{std}}$ two diagonals, $(\tilde{B}_{\text{std}})_{\ell, \ell} = c_\ell$, $(\tilde{B}_{\text{std}})_{\ell, \ell+2} = -1$.

B.2 The normalised basis

The construction of Section 3.3 works in the normalised basis $\varphi_0 = T_0/2^{n/2}$, $\varphi_\ell = T_\ell/2^{(n-1)/2}$ ($\ell \geq 1$), orthonormal under the discrete N -point Chebyshev inner product $\langle T_k, T_\ell \rangle_N = \frac{N}{2} c_\ell \delta_{k\ell}$ [28]: since $N = 2^n$, this gives $\|\varphi_0\|_N = \|\varphi_\ell\|_N = 1$ exactly, so the coefficient vector of a unit-norm function is a unit vector. Writing $S = \text{diag}(\sqrt{c_\ell})$, the normalised factors are $\tilde{B} = S \tilde{B}_{\text{std}} S^{-1}$ and $\tilde{D} = S \tilde{D}_{\text{std}} S^{-1}$, that is

$$\tilde{D}_{\ell, \ell+1} = 2(\ell + 1)\sqrt{c_\ell/c_{\ell+1}}, \quad \tilde{B}_{\ell, \ell} = c_\ell, \quad \tilde{B}_{\ell, \ell+2} = -\sqrt{c_\ell/c_{\ell+2}}, \quad (\text{B.2})$$

so $\tilde{D}$ has the single non-zero diagonal $(2\sqrt{2}, 4, 6, 8, \dots)$ and $\tilde{B}$ has main diagonal $(2, 1, 1, \dots)$ and second superdiagonal $(-\sqrt{2}, -1, -1, \dots)$. Both hold $\mathcal{O}(N)$ non-zero entries at any N . The dense Chebyshev derivative matrix is their exact quotient,

$$G = \tilde{B}^{-1}\tilde{D}, \quad (\text{B.3})$$

an algebraic identity rather than a numerical approximation: $\tilde{B}$ and $\tilde{D}$ are read directly off the same recurrence that defines G (Equation (B.1)), so the equality holds exactly at every N , independent of whichever values a test suite happens to check. Appendix F reports this identity's residual against the reference implementation, alongside the rest of the composition-algebra checks.

B.3 The Ultraspherical Connection

The banded factorisation above is the first-derivative case of the ultraspherical spectral method of Olver and Townsend [2]. Stating the correspondence precisely is what fixes what this thesis adds to it.

Olver and Townsend represent a function's derivative not in the Chebyshev basis T_k but in the ultraspherical (Gegenbauer) basis $C_k^{(\lambda)}$, the family of polynomials orthogonal with respect to the weight $(1 - x^2)^{\lambda-1/2}$; at $\lambda = 1$ these are the Chebyshev polynomials of the second kind, $C_k^{(1)} = U_k$. Differentiation of a Chebyshev series obeys

$$\frac{dT_k}{dx} = k C_{k-1}^{(1)}, \quad k \geq 1 \quad (\text{B.4})$$

[2, Eq. 2.5], so for $u = \sum_k u_k T_k$ the coefficients of u' in the $C^{(1)}$ basis are read off directly: $(\mathcal{D}_0 u)_k = (k+1)u_{k+1}$, a single superdiagonal, sparse at every truncation. Because u' 's coefficients now live in a different basis to u 's own, a second, purely algebraic operator is needed to convert a function's own Chebyshev coefficients into the $C^{(1)}$ basis without differentiating it: $\mathcal{S}_0$, with $(\mathcal{S}_0)_{k,k} = 1$ ($k = 0$), $\frac{1}{2}$ ($k \geq 1$), and $(\mathcal{S}_0)_{k,k+2} = -\frac{1}{2}$ [2, Eq. 2.8]. Comparing entries, $\tilde{D}_{\text{std}} = 2\mathcal{D}_0$ and $\tilde{B}_{\text{std}} = 2\mathcal{S}_0$ exactly: the recurrence $\tilde{B}_{\text{std}} p = \tilde{D}_{\text{std}} a$ of Equation (B.1) is literally $\mathcal{S}_0 p = \mathcal{D}_0 a$, doubled by the convention that folds the weight of T_0 into $c_0 = 2$ rather than into the polynomial itself. This is the discrete shadow the framing of Section 3.3 claims: the descriptor recurrence does not compute a new identity, it asserts an existing one – that p 's own Chebyshev coefficients, converted into the $C^{(1)}$ basis by $\mathcal{S}_0$, must equal the derivative computed directly by $\mathcal{D}_0$ – as one row of a linear system, rather than solving it for p via $\mathcal{S}_0^{-1}$.

The generalisation to order $k \geq 1$ in [2] raises the basis one level per derivative: a λ -th derivative needs coefficients in $C^{(\lambda)}$, obtained from the analogous pair $dC_k^{(\lambda)}/dx = 2\lambda C_{k-1}^{(\lambda+1)}$ ($k \geq 1$, giving $\mathcal{D}_\lambda$, still a single superdiagonal) and a conversion operator $\mathcal{S}_\lambda : C^{(\lambda)} \rightarrow C^{(\lambda+1)}$, still two diagonals [2, Eqs. 3.2–3.3]. To assemble one equation of order k , every term is raised to the common top basis $C^{(k)}$ by a chain $\mathcal{S}_{k-1} \cdots \mathcal{S}_{\lambda+1}$ applied after each term's own $\mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$, giving one almost-banded operator $L = \mathcal{M}_k[a^k]\mathcal{D}_k + \sum_{\lambda=0}^{k-1} \mathcal{S}_{k-1} \cdots \mathcal{S}_\lambda \mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$ that is solved once for the coefficients of $u^{(k)}$ [2, §3.2] – so a k -th order equation needs k distinct conversion operators $\mathcal{S}_0, \dots, \mathcal{S}_{k-1}$ chained together and the lower-order unknowns eliminated.

The descriptor construction of Section 3.3 takes a different route to the same sparsity. Rather than raising the basis and eliminating, it keeps every derivative's coefficients in the *same* normalised Chebyshev basis, as its own block $c^{(j)}$ of w , and reuses the single $\lambda = 0$ pair $(\tilde{B}, \tilde{D})$ between every consecutive pair of blocks (Equation (3.6)). No elimination is performed, and no operator beyond $\tilde{B}$, $\tilde{D}$ and the multiplication operators of Appendix B.5 is required at any order k – the price is the larger state w that carries every intermediate derivative explicitly instead of eliminating it (Appendix B.5). What this thesis adds to [2] is not a new sparsification of differentiation – that is their result – but applying the *same* $\lambda = 0$ identity block-wise to build an unreduced, differential-algebraic residual for the physics-informed Hamiltonian of [1],

whose block structure is exactly what Section 3.4 block-encodes.

B.4 The $N = 4$ instance

At $n = 2$ ($N = 4$), Equation (B.2) gives

$$\tilde{B} = \begin{pmatrix} 2 & 0 & -\sqrt{2} & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \tilde{D} = \begin{pmatrix} 0 & 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 0 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{B.5})$$

The top-right entry $3\sqrt{2}$ of G is the fill-in: coefficient 0 of the derivative depends on coefficient 3 of the input. Row ℓ of G is non-zero at columns $\ell + 1, \ell + 3, \dots$, so this is the extreme case – the lowest-order coefficient can depend on up to half the others, while the highest-order one (row 3 above) depends on none. The factorisation removes the fill-in entirely: no row of $\tilde{B}$ or $\tilde{D}$ touches more than two coefficients.

For the first-order equation $f' + 2f = 0$ of Section 3.3, $w = (c^{(0)}; c^{(1)})$ and $A = \begin{pmatrix} -\tilde{D} & \tilde{B} \\ 2I_4 & I_4 \end{pmatrix}$ is the 8×8 matrix

$$A = \left[\begin{array}{cccc|cccc} 0 & -2\sqrt{2} & 0 & 0 & 2 & 0 & -\sqrt{2} & 0 \\ 0 & 0 & -4 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{array} \right]. \quad (\text{B.6})$$

The top four rows are the recurrence block $-\tilde{D}c^{(0)} + \tilde{B}c^{(1)} = 0$; the bottom four are the differential row $2c^{(0)} + c^{(1)} = 0$. Every non-zero entry lies on a diagonal or a near-diagonal, in place of the dense $G + 2I$ of the eliminated form.

B.5 Higher order, variable coefficients, and the padded block count

For an order- k equation the state is $w = (c^{(0)}; \dots; c^{(k)})$ and A stacks k recurrence block-rows $-\tilde{D}c^{(j)} + \tilde{B}c^{(j+1)} = 0$ above the single differential row $\sum_j M_{a_j} c^{(j)} = 0$: $k + 1$ block-rows, each N coefficients wide, giving the classical $(k + 1)N \times (k + 1)N$ square matrix of Definition 10.

A variable coefficient $a_j(x)$ enters through $M_{a_j} = \sum_q c_q M_{x^q}$, $0 \leq q \leq N$, where c_q are the *monomial* coefficients of a_j : supplied directly as an array $[c_0, c_1, \dots]$, or, when a_j is given as a function, obtained by a least-squares fit to degree $N - 1$ in the Chebyshev basis followed by an exact Chebyshev-to-monomial change of basis. The Chebyshev fit is what pins q to the same resolution N already fixed for the equation; its degree, not the subsequent change of basis, is

the only place a_j 's representation is truncated. M_{x^q} is the resolution- n , square multiplication-by- x^q operator built from $xT_\ell = \frac{1}{2}(T_{\ell+1} + T_{\ell-1})$ with the T_N term dropped. M_{x^q} is banded with bandwidth exactly q , so it folds into the same shift-and-diagonal form as $\tilde{B}$ and $\tilde{D}$. The dropped T_N term is the only approximation in the linear construction: for a smooth product resolved below degree $N - q$ the resulting error is confined to the top $\sim q$ Chebyshev modes and measures $\sim 10^{-13}$.

Padding to a power of two. The classical $(k+1)N \times (k+1)N$ matrix above is exact and complete as a linear system; it is not, however, the object any later stage of the pipeline actually builds. A gate-level realisation must index which block $c^{(j)}$ a given operator acts on with a qubit register, and a register of m qubits addresses 2^m blocks, not $k+1$ of them for an arbitrary k . A is therefore *embedded* in a $2^{\lceil \log_2(k+1) \rceil} N$ -dimensional space: the $k+1$ block-rows and block-columns above are kept exactly as derived, and $2^{\lceil \log_2(k+1) \rceil} - (k+1)$ further diagonal blocks are appended, each an identity I_N pinning its own dummy field to zero, $c_{\text{dummy}}^{(j)} = 0$. This padding is exact: the physical solution already satisfies the dummy rows trivially, since it has nothing in those blocks, so no error is introduced beyond the M_{x^q} truncation above. The qubit cost, however, is not uniform in k :

Order k	Blocks $k+1$	Field qubits $\lceil \log_2(k+1) \rceil$	Padded blocks $2^{\lceil \log_2(k+1) \rceil}$	Dummy blocks
1	2	1	2	0
2	3	2	4	1
3	4	2	4	0
4	5	3	8	3
7	8	3	8	0

so the state w the gate-level construction actually carries has $2^{\lceil \log_2(k+1) \rceil} N$ components, not $(k+1)N$: equal to the unpadded count exactly when $k+1$ is itself a power of two (orders 1, 3, 7, ...), and up to nearly double it otherwise – order 2, the most common case after order 1, already pays for a field register it only three-quarters uses. This is the qubit cost that must always accompany the gate-count comparison of Section 3.6.

- C The Descriptor Construction in Full
- D Proof of the Cost Proposition
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